2022秋高数上期中试题(回忆版).

1. Multiple Choice Questions: (only one correct answer for each of the following questions.)

(1) Which of the following functions is differentiable at $x=0$?

(A) $|x|\sqrt{\sin x+2}$ (B) $|x|+\sqrt{\sin x+2}$ (C) $|x|\sin x$ (D) $|x|+\sin x$

(2) Which of the following functions has an oblique asymptote?

(A) $\frac{\sqrt{2x^3+x+1}}{x+1}$ (B) $\frac{x^4+1}{x^3+\sin x}$ (C) $x+\sin x$ (D) $x+\frac{1}{2+\sin x}$

(3) $\int_{\frac{\pi}{6}}^{\frac{\pi}{2}}\left(\frac{\cos x}{\sin^2 x}+\cos x\right)dx=$

(A) $\frac{3}{2}$ (B) $\frac{2}{3}$ (C) $-\frac{3}{2}$ (D) $-\frac{2}{3}$

(4) Let $f(x)=\begin{cases}\frac{1-\cos x}{x}, & x>0\\ x\sin\frac{1}{x-1}, & x\le 0\end{cases}$

Which of the following must be true?

(A) $\lim\limits_{x\to 0}f(x)$ does not exist.

(B) $\lim\limits_{x\to 0}f(x)$ exists and f is not continuous at $x=0$.

(C) f is continuous at $x=0$ and f is not differentiable at $x=0$.

(D) f is differentiable at $x=0$.

(5) If there is a jump discontinuity for the function $y=f(x)$ at $x=0$, then which of the following limits must exist?

(A) $\lim\limits_{x\to 0}f(x^2)$ (B) $\lim\limits_{x\to 0}(f(x))^2$ (C) $\lim\limits_{x\to 0}f(x^3)$ (D) $\lim\limits_{x\to 0}(f(x)-f(-x))$

2. Fill in the blanks.

(1) $\lim\limits_{n\to\infty}\frac{1}{n}\left(\sin\frac{\pi}{n}+\sin\frac{2\pi}{n}+\cdots+\sin\frac{n\pi}{n}\right)=$ _____.

(2) If the line $y=9x+b$ is a tangent line of the curve $y=x^3-3x$, then $b=$ _____.

(3) If $f(x)=\sqrt{x\sqrt{x+\sqrt{x}}}$, then $f'(1)=$ _____.

(4) $\lim\limits_{x\to 0}\frac{x\tan x}{1-\cos x}=$ _____.

(5) Let $f(x)=\tan x$. Then $f^{(4)}(0)=$ _____.

3. Prove that there is only one real root for the equation $x^5+2x-100=0$.

4. Compute $\int_0^1(1+x)^2(1-x)^5dx$

5. Find the linear approximation of $f(x)=\frac{2}{1-x}+\sqrt{1+x}$ at $x=0$.

6. Find the contants a and b such that the function $f(x)=\begin{cases}\frac{2x^2-x+b}{x-1}, & x>1\\ a, & x\le 1\end{cases}$ is continuous at $x=1$.

7. Let $f(x)=\frac{x^3}{2(x-1)^2}$

(a) Identify the inflection points and local maxima and minima of the function that may exist.

(b) Identify the horizontal, vertical and oblique asymptotes that may exist.

(c) Sketch the graph.

8. Let $y=f(x)$ be an implicit function defined by the equation $2y^3-y^2+3xy-2x^2-2=0$. Find the equation of the tangent line to the curve $y=f(x)$ at $x=1$.

9. Let f be continuous on $(-\infty,\infty)$ and define $F(x)=\int_0^{x^2} xt\,f(x^2-t^2)\,dt$. Find $F'(x)$.